

AeroBook Flight Platform

Online Airline Ticket Booking & Management System

PRESENTED BY

Amir Sohel

Software Engineer

- * Project Role: Full-Stack & Backend Engineering
- * Designed Modular Microservices Architecture
- * Built Spring Boot REST APIs & MySQL Databases
- * Implemented Flight Booking & Web Check-In Workflows
- * Complete Test Automation (109 Tests) & Documentation

PROJECT HIGHLIGHTS

What is AeroBook:

- * A web application for booking flights and web check-in.

Architecture Paradigm:

- * Microservices Architecture (Independent services).

Core Technologies:

- * Java 17, Spring Boot, Spring Cloud, MySQL, Docker.

Key Focus:

- * Safe, simple, fast ticket reservations with zero overselling.

Testing Status:

- * 100% Tests Passed (109/109) with 50 REST API Endpoints.

Why Microservices Instead of One Big App?

Easy Comparison Between Traditional Monoliths and AeroBook Microservices

Traditional Monolith (One Big App)

* Single Point of Failure:

Everything is inside one program. If one small feature crashes, the whole website stops working.

* Shared Database Slowdowns:

All features share one single database. When thousands search for flights, it slows down logins and payments.

* Hard to Update:

Even fixing a small bug requires stopping and restarting the entire system.

* Difficult Teamwork:

Multiple developers working in one giant codebase often create code conflicts.

AeroBook Microservices (Our Approach)

* Independent Services:

Divided into small separate services. If check-in is busy, flight search and booking still run smoothly.

* Separate Databases:

Each service has its own private database, so search traffic does not lock up booking tables.

* Easy Updates:

We can update or fix one service without shutting down the other services.

* Fast and Scalable:

If flight bookings spike during holidays, we can easily run extra copies of just the booking service.

The Core Microservices in AeroBook

Each Service Does Exactly One Job Well (Simple & Modular)

1. API Gateway (Port 8083)**The Main Entrance**

Front door for all requests; automatically sends requests to the right service.

2. Service Discovery (Port 8761)**The Phonebook**

Netflix Eureka keeps a live list of all running microservices automatically.

3. Auth Service (Port 8082)**Security & Login**

Handles user registration, password encryption (BCrypt), and login tokens (JWT).

4. User Service (Port 8084)**Customer Profiles**

Stores customer details like name, email, phone number, and nationality.

5. Flight Service (Port 8087)**Flight Schedules**

Manages airplanes, flight schedules, departure times, and available seats.

6. Fare Service (Port 8089)**Ticket Pricing**

Calculates Economy, Business, and First Class ticket prices plus taxes.

7. Booking Service (Port 8088)**Reservation Engine**

Creates flight bookings, saves passenger details, and generates PNR codes.

8. Check-In Service (Port 8090)**Airport Web Check-In**

Lets passengers choose their seat (e.g. 12A) and get a digital boarding pass.

How AeroBook Works: Step-by-Step

A Simple Walkthrough of the Customer Journey from Search to Boarding Pass

Step 1**Search Flights**

Passenger enters origin city, destination, and travel date. System displays all available flights with live ticket prices.

Step 2**Select Flight & Add Passengers**

Passenger picks a flight, selects Economy or Business class, and enters passenger names and ages.

Step 3**Seat Check (No Overbooking)**

Before confirming, Flight Service safely checks and decrements available seats so two people cannot buy the same seat.

Step 4**Instant PNR Generation**

Booking Service creates the reservation, confirms the ticket, and generates a unique PNR code (e.g. PNR-AB789).

Step 5**Web Check-In (T-24 Hours)**

Within 24 hours of flight departure, the passenger logs in, enters PNR, and chooses their seat (e.g. 12A).

Step 6**Digital QR Boarding Pass**

Check-In Service assigns the seat and immediately creates a digital boarding pass with a QR code for airport gates.

Testing, Achievements & Completed Work

Everything Built and Thoroughly Tested in AeroBook

109 / 109

Automated Tests Passed

All 109 unit tests passed with 0 errors.

50 Endpoints

REST API Endpoints

50 tested API routes with JSON input & output.

6 Databases

Clean Database Storage

Dedicated MySQL databases for clean data separation.

6 Diagrams

Complete Blueprint Suite

Architecture, Use Case, Class, DFD, ER & Activity.

CORE PRACTICAL FEATURES COMPLETED

• Secure User Login:

Passwords encrypted using BCrypt; secure JWT tokens issued on login.

• Multi-Class Fares:

Automated calculation of Economy, Business, and First Class pricing with 18% GST.

• Web Check-In & Boarding Pass:

Online seat selection with instant digital boarding pass reference and QR code.

• Real-Time Flight Search:

Fast flight searches by city route and date with available seat counts.

• Safe Seat Reservation:

Row-level locking ensures that two users cannot book the same seat at once.

• Structured Logging:

SLF4J and Logback logging implemented for easy debugging and monitoring.

Future Scope: What We Can Add Next

Practical and Realistic Enhancements to Expand the AeroBook Platform

1

Live Payment Gateway Integration

Connect with popular payment gateways like Razorpay, Stripe, or UPI so passengers can make real online payments and get instant refunds on cancellation.

2

Automated Email & SMS Notifications

Automatically send booking confirmation emails with PDF tickets and SMS reminders about flight departure times and gate updates.

3

Interactive Airplane Seat Map in UI

Add a visual airplane cabin map where passengers can see available seats in green and taken seats in red, making seat selection fun and intuitive.

4

Live Flight Status & Delay Tracking

Add real-time flight updates (On Time, Delayed, Gate Changes) and weather notifications for traveling passengers.

5

Mobile App for Android & iOS

Build a lightweight mobile app using Flutter or React Native so passengers can easily book flights and show boarding passes right from their phones.